



View PDF

Download full issue

Biomedical Signal Processing and Control

Volume 55, January 2020, 101624

Automatic and tunable algorithm for EEG artifact removal using wavelet decomposition with applications in predictive modeling during auditory tasks

Nikesh Bajaj^{a, b}JesA's Requena CarriA'n^aFrancesco Bellotti^bRiccardo Berta^aAlesandro De Gloria^b

Show more



Share



Cite

<https://doi.org/10.1016/j.bspc.2019.101624>[Get rights and content](#)

Full text access

Highlights

•

The proposed artifact removal approach is based on Wavelet Packet Decomposition, with three operating modes namely Soft-thresholding, Linear attenuation and Elimination.

•

The proposed approach allows controlling the suppression or removal information with two tuning parameters.

•

Results show the performance of predictive modeling can be improved by properly tuning the parameters.

•

Results show the performance of proposed approach is better in comparison to ICA-based approach and other wavelet-based approach.

Abstract

Brain–computer interface (BCI) systems are becoming increasingly popular nowadays. Electroencephalogram (EEG) signals recorded by BCI systems are however frequently contaminated by artifacts and while applying any artifact removal algorithm, precautions should be taken not to remove useful information. Widely and most popular approaches to remove artifacts from EEG are based on independent component analysis (ICA), which relies on the multichannel EEG signal and needs an expert to manually pick the artifactual component to remove it or needs reference signals of artifacts. Recently, wavelet-based approaches have been proposed and demonstrated as well-suited for single channel EEG signal. However, control over the loss of information still remains an issue. Therefore, in this paper, we propose an algorithm based on wavelet packet decomposition (WPD) that allows controlling the suppression or removal of presumed artifacts, by tuning intuitive parameters. The proposed algorithm has three operating modes and two tuning parameters. We study the performance of the proposed algorithm and compare it with ICA-based approaches and a comparative wavelet-based approach on an EEG dataset collected for a study of auditory tasks. In addition to visual inspection, spectral response and distribution, the results of predictive tasks show that the proposed approach performs better than ICA-based approach and performance can be further improved by properly tuning the parameters for an individual predictive model. The proposed approach also performs better in terms of retention of neural information and removal of artifactual noise measured by mutual information and correlation coefficient, as compared to the comparative wavelet-based approach.



Previous

Next



Keywords

Artifacts removal; Auditory task; Electroencephalogram; Wavelet packet decomposition; Brain–computer interface; Predictive modeling

Cited by (0)

[View Abstract](#)

Â© 2019 Elsevier Ltd. All rights reserved.



About ScienceDirect



Remote access

Advertise



Contact and support



Terms and conditions



Privacy policy



Cookies are used by this site. **Cookie Settings**

All content on this site: Copyright Â© 2025 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

